

Introduction to Digital Forensics



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Digital Forensics

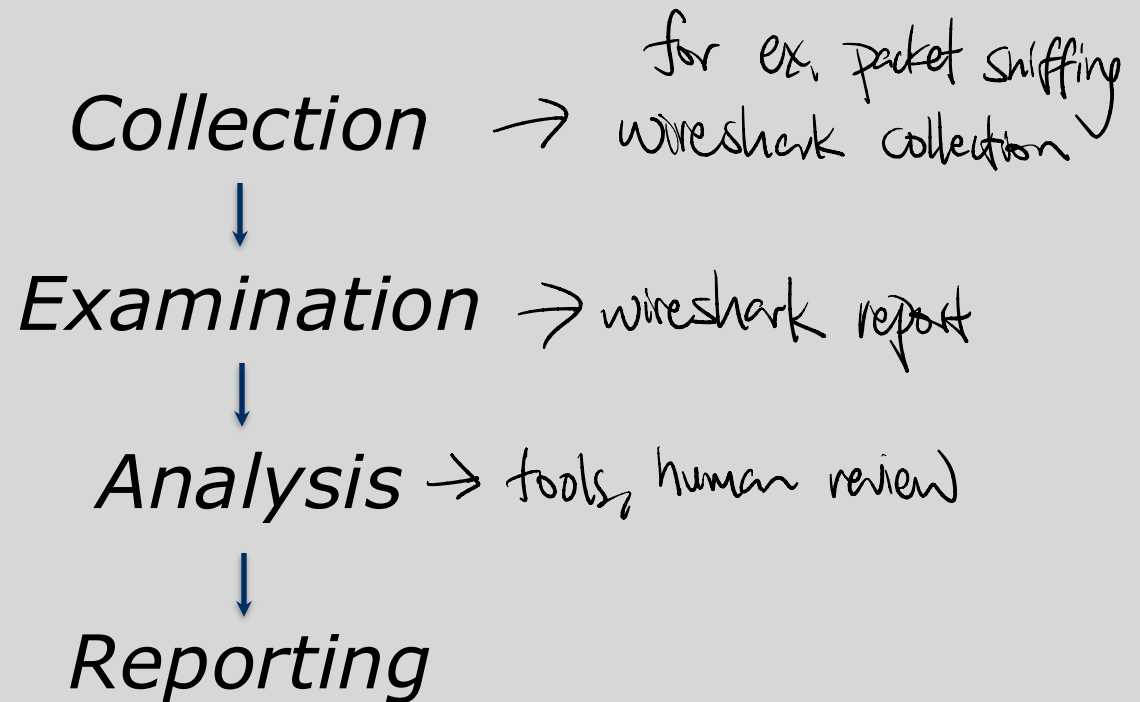
*"Cyber Forensics is the acquisition, preservation and analysis of Electronically Stored Information (ESI) in such a way that ensures its admissibility for use as either evidence, exhibits, or demonstratives in a **court of law**."*

-- Scott Ellis, Kcura Corporation

*"[Digital Forensics] is considered the application of science to the identification, collection, examination, and analysis of data while preserving the **integrity of the information** and maintaining a **strict chain of custody** for the data."*

-- NIST, Integrating Forensic Techniques into Incident Response

Forensic Procedures



Forensic Procedures

Collection → *Examination* → *Analysis* → *Reporting*

Notification and Authorization

- Data Owner
- Legal Issues
- Objectives

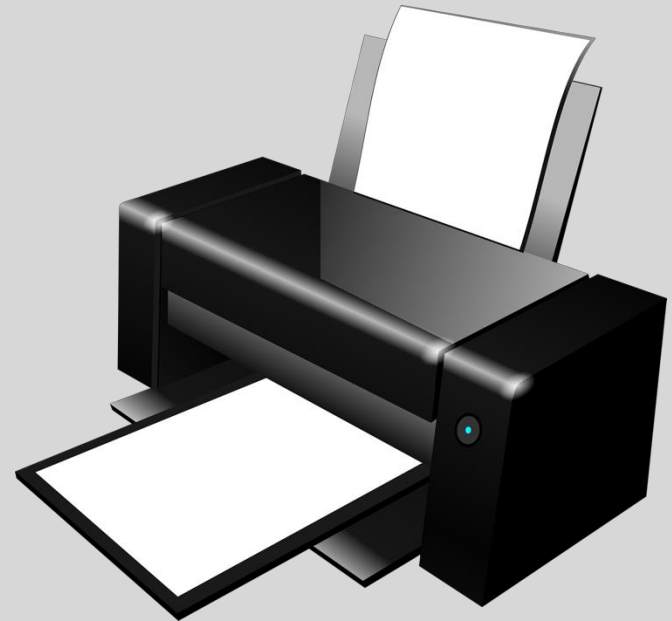


Forensic Procedures

Collection → *Examination* → *Analysis* → *Reporting*

Identifying Data Sources

- Hard Drives / Memory
- External Devices
- Cloud Data



Forensic Procedures

Collection → *Examination* → *Analysis* → *Reporting*

Order of Acquisition

2

RAM / Memory
gone after shutdown

4

External Storage
Devices
small, easy to get

Effort vs. Value



1

hard to get
CPU Registers /
Cache

3

Hard Drive
easy as 4, f6h

note: volatility

Forensic Procedures

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Extracting Evidence

- “Image” Device
- Always work on a copy

talked about next slide



Forensic Images

Forensic Images - a **bit-by-bit exact** copy of the memory space of the device to be examined.

↳ avoids accidentally tampering w/ original data (bad!)

Hash Values - a unique numeric value of fixed length acquired through the application of a mathematical algorithm on digital data, used for ensuring data integrity.

↳ identical data produces same hash

Forensic Procedures

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Tools Needed

- Write Blockers
 - ↳ hw ones preferred, OS-agnostic
- Image Analyzers
 - ↳ product FTK Imager Pro, for ex.
- Hex Editors
 - ↳ bit-by-bit (almost) viewers

Tool Validation



<https://www.logicube.com/shop/writeprotect-desktop/>

Forensic Tools

Write Blockers - devices that are used to stop any changes being made to the original device when making a forensic image.

Image Analyzers – software that previews, organizes and/or extracts data to allow for examination of digital artifacts originating from forensic images.

Forensic Tools

Forensics Tools – digital forensic tools created by software developers for in-depth investigations.

- Disk and Data Capture Tools
- File Viewers
- File, Registry, Internet and Email Analysis
- Mobile Devices Analysis Tools
- Network Forensics Tools

Tool Validation – the **verification** of the software and technology used to perform a digital forensics investigation.

- Hashing to Validate Results
- Multiple Tools or Techniques
- Manual Carving for Control or Validation
- Forensics Tool Testing Programs

Forensic Procedures

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Analysis

- Artifacts and Devices
- Relevant Information

- **Timeline**

↳ piece together to form a case

- Process and Methods

↳ e.g. making logs, document everything

- **Integrity of Data**

↳ how do ppl know I didn't tamper w/ it?



Forensic Artifacts

Forensic Artifacts - **digital evidence**, an object of digital archeological interest.

- Documents
- Contact Lists, Emails
- Calendars
- Browser History, Cookies, and Bookmarks
- **Pictures, Images, and Digital Photos**
- Audio and Video Files
- Thumbnails
- Hidden, System and Temporary Files

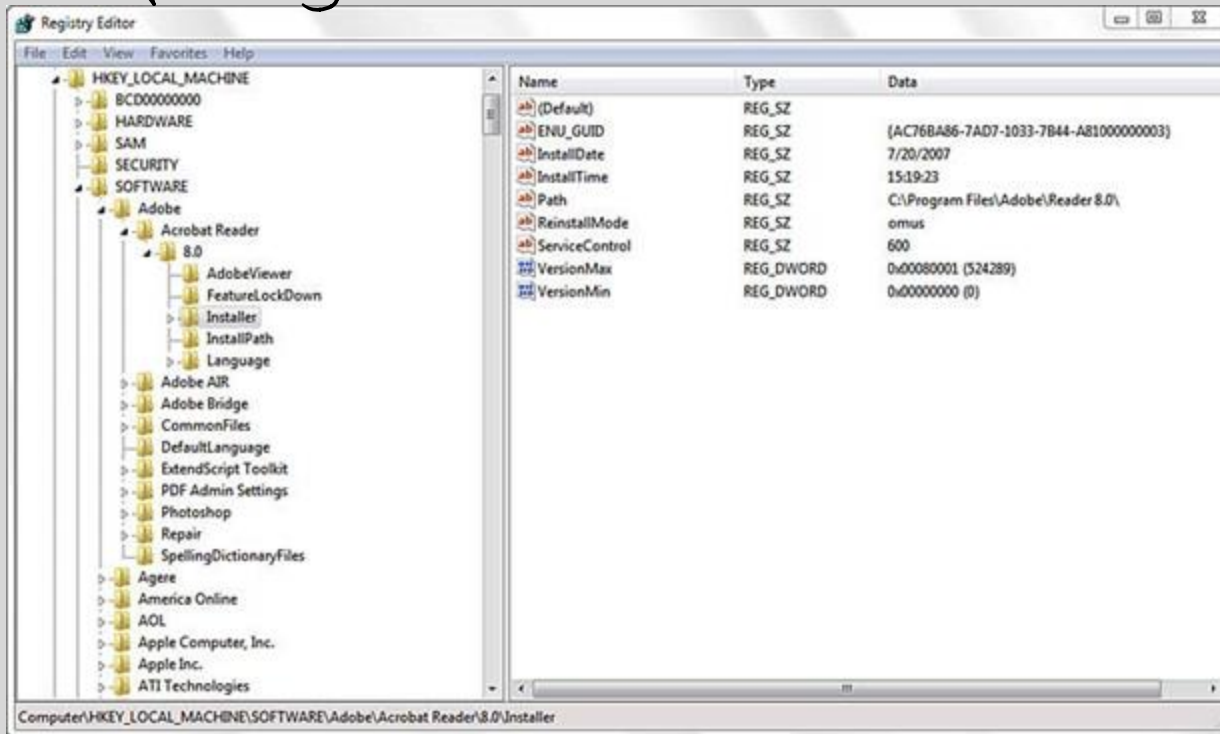
Other Data Sources

- [Mobile Devices]
- [Virtual Machines]

Registry

Registry - A central hierarchical **database** used in Microsoft Windows to store information that is necessary to configure the system for one or more users, applications and hardware devices.

may show previously installed software



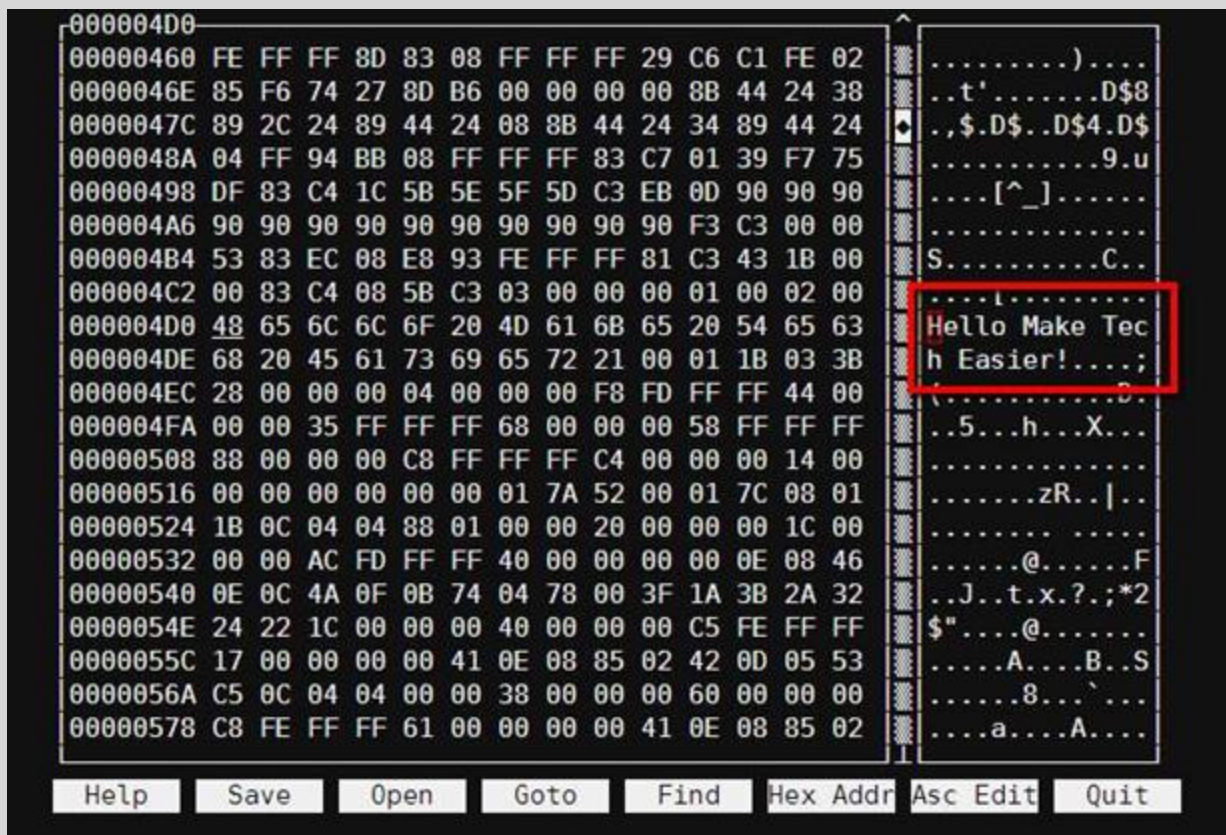
Event Logs

Event Logs - log files are special files that record significant events, e.g. logins or program errors. *unauthorized access attempts?*

Security Number of events: 17 (!) New events available				
Keywords	Date and Time	Source	Event ID	Task Category
 Audit Success	3/9/2011 7:41:16 AM	Microsoft Windo...	4624	Logon
 Audit Success	3/9/2011 7:40:56 AM	Microsoft Windo...	5136	Directory Service Changes
 Audit Success	3/9/2011 7:40:29 AM	Microsoft Windo...	4624	Logon
 Audit Success	3/9/2011 7:40:25 AM	Microsoft Windo...	4624	Logon
 Audit Success	3/9/2011 7:40:25 AM	Microsoft Windo...	4624	Logon
 Audit Success	3/9/2011 7:40:25 AM	Microsoft Windo...	4624	Logon
 Audit Success	3/9/2011 7:40:25 AM	Microsoft Windo...	4624	Logon
 Audit Success	3/9/2011 7:40:07 AM	Microsoft Windo...	4688	Process Creation
 Audit Success	3/9/2011 7:39:29 AM	Microsoft Windo...	4624	Logon
 Audit Success	3/9/2011 7:38:29 AM	Microsoft Windo...	4624	Logon
 Audit Success	3/9/2011 7:38:00 AM	Microsoft Windo...	4624	Logon
 Audit Success	3/9/2011 7:37:29 AM	Microsoft Windo...	4624	Logon
 Audit Success	3/9/2011 7:37:26 AM	Microsoft Windo...	4689	Process Termination
 Audit Success	3/9/2011 7:37:26 AM	Microsoft Windo...	4688	Process Creation
 Audit Success	3/9/2011 7:37:11 AM	Microsoft Windo...	4624	Logon
 Audit Success	3/9/2011 7:37:11 AM	Microsoft Windo...	4624	Logon

Hex Editor

Hex Editor - A hex editor is a software tool used to directly view and modify the binary data (shown in hexadecimal format) of files.



File Carving

File Carving - a technique used to recover files or fragments of files when corrupted, missing or purposely deleted.

JPEG

Header: ffd8ffe00010

Footer: ffd9

FFD8FFE000104A464946000102010048
00480000FFE11B344578696600004D4D
002A000000008000A010F

...

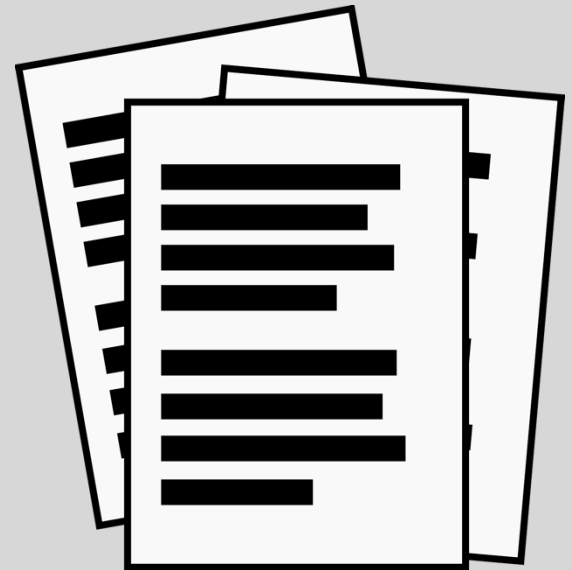
F2B54840253BA4AA67F932C6EE14C445
5991B9E2C18FC66BAED9919934BCC4A1
3AB86CE14B7FFFD9

Forensic Procedures

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Reporting

- Summary
- Scope and Objectives
- Examination Results and Details
- **Conclusion**
- Supporting Documents



What are Metadata and ExifTool?

- *Metadata* – data that provides information about other data, useful to find information about file creation, access, and author.
- *MAC* – file metadata containing Modified, Access, and Creation dates and times. *Useful for checking if edited*
- *ExifTool* – a command-line interface used for reading, writing, and editing metadata.
 - ❖ For our exercise, will only utilize the reading feature of Exiftool

Discussion Questions

- Is digital photo metadata useful as evidence? Is it viable/absolute? What would make it useful or not useful?

↳ can be to establish a timeline

- Can the fields in the metadata be altered? Could we tell if the metadata in a photo had been altered?

↳ yes? idk? other fields may be a clue. Also analyze photo itself for inconsistencies

- Are there privacy concerns? What are they?

↳ if I can see metadata, so can places I upload it to (Instagram, iMessage)

- What should be communicated to students about photo metadata? How can this be applied to the classroom?

↳ settings on camera that can omit certain fields, etc.

References

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